CHE629 FINAL PROJECT
EFFICIENT DRUG-TARGET AFFINITY PREDICTION

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ABSTRACT

Drug–Target Affinity (DTA) prediction is important in drug discovery as it helps estimate how strongly a drug binds to a target protein, while reducing the need for slow and costly experiments. In this work, we study a deep learning model based on the EDTA architecture, which combines CNN, BiLSTM and attention mechanisms in a parallel structure to capture both local and long-range patterns. The model uses different representations of drugs and proteins to learn structural and sequential information effectively. It is implemented in PyTorch and evaluated on the Davis dataset, showing good prediction accuracy with efficient computation, making it suitable for large-scale drug screening.

1 INTRODUCTION

Drug discovery is a complex and time-consuming process that often requires many years of research and significant financial investment. One of the key steps in this process is identifying how strongly a drug molecule interacts with a target protein, commonly referred to as drug–target affinity (DTA). Accurate prediction of DTA helps in selecting promising drug candidates at an early stage, reducing the need for expensive laboratory experiments.

To overcome these limitations, computational approaches have been developed to estimate binding affinity using machine learning and deep learning techniques. Early machine learning models relied heavily on handcrafted features, which limited their ability to generalize across different datasets.

With the advancement of deep learning, models such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Transformers have been applied to DTA prediction. CNNs are effective in capturing local structural patterns, while RNN-based models like LSTM are useful for learning sequential dependencies. However, many existing approaches either combine these models in a sequential manner or rely on complex graph-based architectures (GNNs), which can increase computational cost and sometimes lose important sequence information.

In this work, we focus on an efficient deep learning approach inspired by the EDTA model, which combines CNN, Bidirectional LSTM (BiLSTM), and attention mechanisms in a parallel structure. This design allows the model to capture both local and long-range interactions without significantly increasing computational complexity. The use of multiple feature representations for drugs and targets further improves the model's ability to learn meaningful patterns.

The objective of this study is to implement and analyze this approach using standard benchmark datasets such as Davis, and evaluate its performance using commonly used metrics. The results demonstrate that a well-designed and efficient model can achieve strong prediction accuracy while remaining computationally practical.

1.1 FEATURE REPRESENTATION

In the study, Drug–target pairs are represented using multiple features. The drug is described using two types of features, while the target is represented using three different features. These five features together form twenty-one possible combinations of drug–target representations. Each combination is evaluated by developing separate deep learning model variants with an attention mechanism.

Among the twenty-one feature combinations evaluated on the Davis and KIBA datasets, the combination of drug SMILES, drug features, target features, Byte Pair Encoding (BPE) for targets and potential residue features was found to be the most effective for drug–target affinity prediction.

2 Davis Dataset

In this study, the Davis dataset is used for training and evaluation. The Davis dataset is a widely used benchmark dataset for drug–target affinity prediction. It consists of interactions between 68 drug molecules and 442 protein targets, resulting in a total of 30,056 drug–target pairs. The binding affinity between each drug–target pair is measured using the dissociation constant (K_d), which indicates the strength of interaction between the drug and the target protein.

Since the K_d values vary over a large range, they are commonly transformed into pK_d values using a logarithmic scale for better numerical stability and model performance.

2.1 Cold Start Setting

The cold-start problem occurs when the model is tested on drugs or targets that were not seen during training. This is an important challenge in real-world drug discovery. There are three cases: drug cold-start (new drug, known target), target cold-start (known drug, new target), and drug–target cold-start (both drug and target are new). To evaluate this, the Davis dataset is split in a way that follows existing methods for fair comparison. The model is then tested on these unseen combinations.

3 Input Features

The drug is represented using two features: SMILES sequences, which describe the chemical structure and atomic-level features that capture physicochemical properties. The target protein is represented using three features: sequence-based target features, Byte Pair Encoding (BPE) to capture frequent subsequences, and potential residue features that highlight important binding regions.

3.1 Drug SMILES Representation

Drug molecules are represented using SMILES (Simplified Molecular Input Line Entry System) strings, which encode the chemical structure of a compound, including its atoms, bonds and ring formations. Each SMILES string is composed of a set of up to 64 unique characters. For model processing, each character is mapped to a corresponding integer value in the range of 1 to 64.

For example, the SMILES string "O=CONC(=O)" is converted into a numerical sequence such as [48, 40, 42, 48, 14, 42, 1, 40, 48, 31]. To ensure uniform input size, all SMILES sequences are standardized to a fixed length of 100. If a sequence exceeds this length, it is truncated to the first 100 characters. If it is shorter, zero-padding is applied to reach the required length.

However, in our implementation, the dataset does not provide a predefined SMILES character encoding. Therefore, all SMILES strings were analyzed to identify the total number of unique characters present. It was found that there are 32 unique characters in the dataset. An additional value, 0, is used for padding, resulting in a total of 33 unique tokens. Thus, instead of using 64 possible characters as mentioned in the original approach, our representation is limited to 33 distinct values.

3.2 Drug Feature Representation

In addition to SMILES strings, drugs are also represented using atomic-level features to capture their structural and physicochemical properties. Analysis of the dataset shows that more than 90% of the drug molecules contain fewer than 50 atoms. Therefore, the maximum number of atoms for each drug is fixed at 50.

Each atom is described using a set of 24 features, which include properties such as atom type, atomic mass, number of bonded neighbors, number of hydrogen atoms, hybridization state and other chemical characteristics like aromaticity, charge and donor/acceptor properties. Based on this,

each drug is represented as a matrix of size 50×24. If a drug contains fewer than 50 atoms, the remaining rows are padded with zeros to maintain a fixed input size.

3.3 Target Feature Representation

Target proteins are represented using sequence-based features derived from their amino acid composition. Each residue is encoded using a combination of three components: a 21-dimensional one-hot vector for amino acid type (including one for unknown residues), a 14-dimensional vector capturing residue properties (such as polar, non-polar, acidic, and basic), and a 7-dimensional vector of physicochemical properties. These are combined to form a 42-dimensional feature vector for each residue.

To ensure uniform input size, each protein sequence is fixed to a maximum length of 500. Sequences longer than 500 are truncated, while shorter sequences are padded with zeros. As a result, each target is represented as a matrix of size 500×42.

3.4 BPE Encoding for target

In addition to sequence-based features, target proteins are also represented using Byte Pair Encoding (BPE) to capture frequently occurring subsequences. BPE helps in learning meaningful patterns by grouping common residue sequences instead of treating each amino acid independently.

In this approach, protein sequences are first broken into smaller subsequences using a predefined dictionary of frequent patterns. These subsequences are then encoded into unique integer values based on their frequency and order. This allows the model to capture both local motifs and global sequence information more effectively. In the notebook implementation, a BPE algorithm is applied to generate sequences, where frequent consecutive subsequences are iteratively merged to form a vocabulary. Each protein sequence is then tokenized using this vocabulary and converted into a sequence of integer indices. Similar to other inputs, these sequences are fixed to a maximum length (400 for Davis Dataset), with truncation and zero-padding applied where necessary.

3.5 Potential residue feature for target

Target proteins also contain specific regions responsible for binding with drugs, known as potential residues. These residues are identified from predicted 3D structural information and represent the most relevant sites for drug interaction.

Each potential residue is described using a set of 67 features, including solvent-accessible surface area (SASA), angular features (phi and psi angles), BLOSUM62 encoding, residue type, secondary structure and other physicochemical properties. Based on this, each target is represented as a matrix of size $L \times 67$, where L is the number of potential residues.

To maintain consistency, a maximum length of 200 residues is used. If the number of potential residues is less than 200, zero-padding is applied. This representation helps the model focus on important binding regions and improves prediction accuracy.

4 Model Architecture

The model consists of three main parts: input, feature extraction, and output. Multiple input features (drug SMILES, drug features, target features, BPE encoding, and potential residue features) are first passed through separate embedding layers to obtain uniform representations.

These embeddings are then processed in parallel using 1D CNN layers to extract local patterns, while the SMILES input is additionally passed through a Bidirectional LSTM with attention to capture sequential dependencies. The outputs from all branches are concatenated and passed through a self-attention layer to highlight important interactions. Finally, the combined representation is fed into multiple fully connected (FC) layers to predict the drug–target affinity.

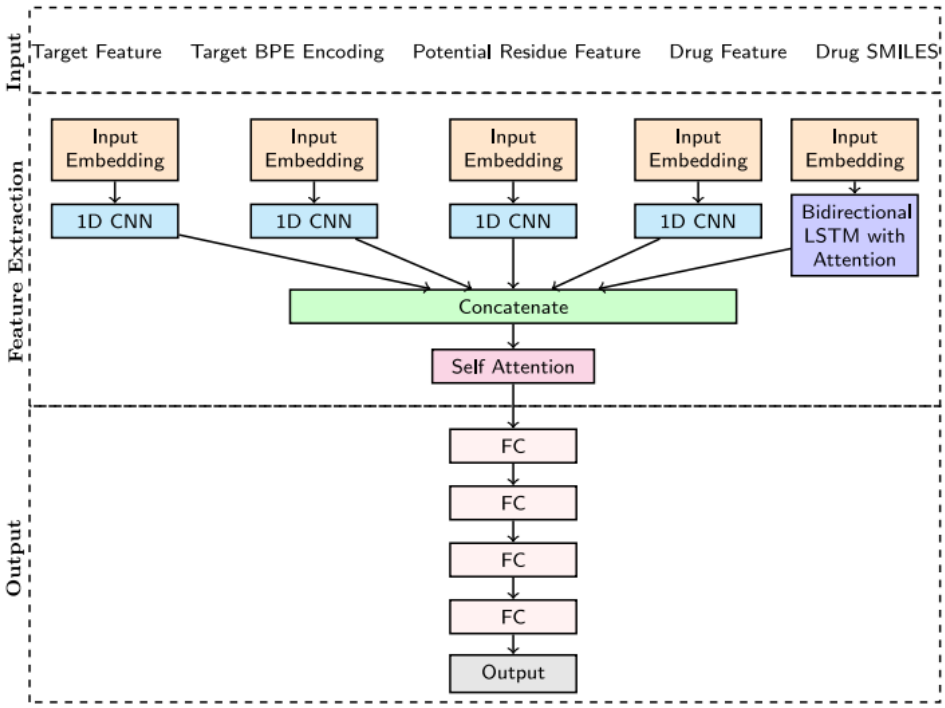

Figure 1: Model architecture of the proposed EDTA-based approach.

4.1 Embedding layer

The embedding layer is the first component of the feature extraction module and is used to convert all input features into a uniform dense representation. Since the input features have different dimensions and formats, the embedding layer maps each of them into a fixed 128-dimensional space.

For each input feature, the embedding layer generates a sequence of vectors, where the length of the sequence remains the same as the original input, but each element is now represented by a 128-dimensional vector. This helps in reducing dimensional differences across features and converts sparse representations into dense ones.

The embedding layer also helps in capturing meaningful relationships between input elements by learning a continuous representation. As a result, all input features are transformed into a consistent format, making them suitable for further processing by CNN and LSTM layers.

4.2 1D CNN Block

The 1D CNN block is applied to each embedded input feature to extract local patterns and contextual information. Each input of shape (B, L, 128) is passed through a series of four 1D convolutional layers with increasing number of filters: 32, 64, 128, and 256. These layers operate along the sequence dimension (L), allowing the model to capture local dependencies between neighboring elements.

Different kernel sizes are used depending on the feature type. For drug features and target BPE encoding, kernel sizes of 4, 6, 6, and 4 are used, while for target features and potential residue features, kernel sizes of 4, 8, 12, and 8 are used. This variation helps in capturing patterns at different scales for different types of inputs. After passing through the CNN layers, each feature is transformed into a higher-level representation with enriched local information. These outputs are then forwarded for further processing and later combined with other feature representations.

4.3 Bidirectional LSTM with Attention

A Bidirectional LSTM (BiLSTM) layer is applied to the embedded drug SMILES representation to capture sequential dependencies in both forward and backward directions. The input to this layer has shape (B, 100, 128) and a two-layer BiLSTM is used to learn both short-term and long-term relationships within the sequence.

The BiLSTM processes the sequence in two directions and produces a hidden representation that captures context from both past and future positions. An attention mechanism is implemented using a linear layer that maps each 256-dimensional hidden state to a scalar score. A masking strategy is applied before the softmax operation to ignore padded positions (where token value is 0), ensuring that attention is only computed over valid tokens. The attention weights are then normalized using softmax. Finally, a weighted sum of the BiLSTM outputs is computed using batch matrix multiplication, producing a single context vector of size (B, 256) for each input sequence

Note: The outputs from all feature extraction branches are combined using a concatenation layer. The concatenated representation results in a feature vector of dimension 1280, which integrates information from all input features for further processing to self-attention layer.

4.4 Self-Attention Layer

The self-attention layer is applied after concatenation of all feature representations, where the input has a dimension of $(B, 1280)$. To reduce computational cost, a bottleneck attention mechanism is used, where the input is first projected to a lower dimension of 128 for query (Q), key (K), and value (V) representations using three linear layers. This reduces the parameter complexity from approximately 6.5 million to 0.65 million parameters.

The projected matrices $Q, K, V \in \mathbb{R}^{B\times 1\times 128}$ are used to compute scaled dot-product attention. The attention score is calculated as softmax, followed by a softmax operation to obtain attention weights. A dropout of 0.1 is applied to these weights for regularization. The attended output is then computed as a weighted sum of V, resulting in a tensor of shape $(B, 1, 128)$, which is further projected back to the original dimension $(B, 1, 1280)$.

Finally, a residual connection is applied by adding the original input to the attended output, followed by layer normalization. The final output of the layer has shape $(B, 1280)$, which preserves the original dimensionality while enhancing important feature interactions.

4.5 Fully Connected Layer

The output from the self-attention layer, of shape (B, 1280), is passed through a series of fully connected (FC) layers to perform the final prediction. The FC block consists of four layers with decreasing dimensions: 512, 256, 128 and 64 neurons, respectively. Each layer applies a linear transformation to progressively reduce the feature dimension while learning higher-level representations.

Non-linear activation functions (such as ReLU) are applied after each layer, along with dropout to prevent overfitting. Finally, an output layer with a single neuron is used to predict the drug–target affinity value, resulting in an output of shape (B, 1). This multi-layer FC structure helps in combining the extracted features and mapping them effectively to a continuous affinity score.

5 Model Training

The model is trained as a regression task using Mean Squared Error (MSE) as the loss function. The Adam optimizer is used for parameter optimization with a learning rate of 0.0002.

Each dataset is randomly divided into six equal partitions, where each partition of the Davis dataset contains approximately 5,010 drug-target pairs. One of the six partitions is reserved as an independent test set. Among the remaining five partitions, four are used for training and one for validation. This partitioning strategy is consistent with that adopted in existing state of the art methods, including 3DProtDTA (Voitsitskyi et al., 2023), DrugMGR (Li et al., 2024), and DeepDTA (Öztürk et al., 2018).

The model is trained for 200 epochs, with performance monitored at each epoch. The model weights corresponding to the lowest validation loss are saved and used for final evaluation on the test set. Training is carried out using mini-batches on a GPU using PyTorch, which allows efficient computation. This setup ensures stable convergence and reduces the risk of overfitting while maintaining good generalization performance.

6 Evaluation Metrics

To evaluate the performance of the model, four commonly used metrics for regression tasks are employed: Mean Squared Error (MSE), Concordance Index (CI), Pearson Correlation Coefficient (PCC), and the Squared Correlation Coefficient (r_m^2).

Mean Squared Error (MSE): MSE measures the average squared difference between the predicted and actual affinity values, and is defined as:

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}\left(Y_{\text{pred}_i} - Y_{\text{act}_i}\right)^2 \tag{1}$$

Concordance Index (CI): CI measures the agreement between predicted and actual values by evaluating how often the predicted ranking of two samples matches their actual ranking. It is defined as:

$$\text{CI} = \frac{1}{Z}\sum_{Y_{\text{act}_i} > Y_{\text{act}_j}} h\left(Y_{\text{pred}_i} - Y_{\text{pred}_j}\right) \tag{2}$$

where Z is a normalization constant and $h(u)$ is a step function that returns 1.0 for $u > 0$, 0.5 for $u = 0$, and 0.0 for $u < 0$. The CI value ranges from 0.5 to 1.0, where 0.5 represents random prediction and 1.0 indicates perfect predictive accuracy. A higher CI value indicates better performance of a model.

Pearson Correlation Coefficient (PCC): PCC quantifies the intensity of the linear relationship between actual and predicted values. It is defined as the covariance divided by the product of their standard deviations:

$$\text{PCC } r = \frac{cov(Y_{\text{act}},\, Y_{\text{pred}})}{\sigma(Y_{\text{act}}) \cdot \sigma(Y_{\text{pred}})} \tag{3}$$

A higher PCC value signifies greater effectiveness of the model.

Squared Correlation Coefficient: The Squared Correlation Coefficient (r_m^2) index evaluates the model's external predictive performance through regression towards the mean. It is calculated using both the predicted and actual values with intercept (r^2) and without intercept (r_0^2), and is defined as:

$$r^2 = \frac{\left|\sum_{i=1}^{N}(Y_{\text{act}_i} - \bar{Y}_{\text{act}})(Y_{\text{pred}_i} - \bar{Y}_{\text{pred}})\right|^2}{\sum_{i=1}^{N}(Y_{\text{act}_i} - \bar{Y}_{\text{act}})^2 \times \sum_{i=1}^{N}(Y_{\text{pred}_i} - \bar{Y}_{\text{pred}})^2} \tag{4}$$

$$\text{SCC}r_m^2 = r^2 \times \left(1 - \sqrt{r^2 - r_0^2}\right) \tag{5}$$

A higher value of r_m^2 signifies better performance of the model.

Together, these metrics provide a comprehensive evaluation of the model by measuring prediction error (MSE), ranking ability (CI), linear correlation (PCC) and external predictive power (r_m^2).

7 Results

7.1 Results on Davis Dataset

The performance obtained in this work is compared with the results reported in the original EDTA paper on the Davis dataset. The paper reports a test MSE of approximately 0.173, CI of 0.909, and

r_m^2 of 0.783. In comparison, our implementation achieves a test MSE of around 0.23, CI of about 0.88, and r_m^2 close to 0.67. While the results are slightly lower than those reported in the paper, the overall trends are consistent, and the model demonstrates comparable performance. The difference in performance may be attributed to variations in implementation details, feature preprocessing, or training setup. Nevertheless, the results confirm that the model is able to capture meaningful patterns in drug–target interactions.

Table 1: 5-Fold Cross-Validation Results on Davis Dataset

Fold	**MSE**	**CI**	**Pearson**	$\mathbf{r_m^2}$
1	0.2324	0.8748	0.8438	0.6793
2	0.2381	0.8819	0.8403	0.6691
3	0.2532	0.8775	0.8345	0.6208
4	0.2380	0.8793	0.8399	0.6826
5	0.2328	0.8839	0.8433	0.6903
Average	**0.2389**	**0.8795**	**0.8404**	**0.6684**
Std Dev	**0.0075**	**0.0032**	**0.0033**	**0.0248**

Table 2: Comparison of our implementation with the state-of-the-art methods on the Davis testing set. Note that bold values indicate the best performance in each column, boxed values indicate the best within either conventional DL or graph-based methods, while shaded values denote the second-best among conventional DL methods.

Methods	**ML/DL**	**Graph/Conventional**	**MSE(↓)**	**CI(↑)**	r_m^2 **(↑)**
KronRLS	ML		0.379	0.871	0.407
SimBoost	ML		0.282	0.872	0.644
GraphDTA-GIN	DL	Graph	0.229	0.893	0.656
MGraphDTA	DL	Graph	0.207	0.900	0.710
WGNNDTA	DL	Graph	0.214	0.892	0.711
3DProtDTA	DL	Graph	0.184	**0.917**	0.722
GDilatedDTA	DL	Graph	0.237	0.885	0.686
DeepDTAGen	DL	Graph	0.214	0.890	0.705
DeepDTA	DL	Conventional	0.261	0.878	0.630
DeepCDA	DL	Conventional	0.248	0.891	0.649
FusionDTA	DL	Conventional	0.208	0.913	0.666
Bicomp-DTA	DL	Conventional	0.237	0.904	0.696
EDTA	DL	Conventional	**0.173**	0.909	**0.783**
EDTA (Our implementation)	DL	Conventional	0.239	0.879	0.668

7.2 Results on KIBA Dataset (Resource-Constrained Evaluation)

The KIBA dataset is significantly larger than the Davis dataset, making it computationally more demanding. Due to hardware and runtime constraints, a full 5-fold evaluation could not be completed. Each training run required approximately 25 hours on a dual T4 GPU setup, whereas the available runtime on Kaggle was limited to 12 hours per session. Experiments were conducted for 2 out of the 5 folds, with first fold trained for 200 epochs and latter one for 110 epochs.

For the evaluated folds, the model achieved a test MSE of approximately 0.19 and a CI of around 0.86, with an r_m^2 value close to 0.71. These results indicate that the model is able to learn meaningful patterns even on a large-scale dataset. In comparison, the original EDTA paper reports a test MSE of 0.145, CI of 0.886, and r_m^2 of 0.787 on the KIBA dataset. While our results are slightly lower,

they are reasonably close considering that only a subset of folds was trained. Overall, the model demonstrates good scalability and maintains competitive performance on larger datasets.

8 Conclusion

In this study, a lightweight conventional deep learning method with attention mechanisms, named EDTA, is implemented and evaluated for drug-target affinity (DTA) prediction. The framework employs multiple explicit feature representations for drugs and targets, integrating four CNN blocks, a BiLSTM, attention mechanisms, and a fully connected neural network (FCNN) to capture both local and long-range interaction patterns.

The results confirm that EDTA effectively learns meaningful drug-target interaction patterns and maintains competitive performance among both conventional DL and graph-based state-of-the-art methods. Its lightweight design requires fewer computational resources, making it suitable for large-scale drug screening. By efficiently narrowing the drug candidate search space, EDTA demonstrates clear potential to accelerate early-stage drug discovery pipelines.

References

[1] Mohamed Abdul Cader, J., Newton, M.A.H., & Sattar, A. (2024). Efficient drug–target affinity prediction via interaction features and parallel CNN–BiLSTM with attention. *School of Information and Communication Technology, Griffith University*. Available at: `https://github.com/nonit-gupta/Drug-Target-Affinity-Prediction/blob/main/drug-aff.pdf`

Dataset

The dataset used in this study are publicly available at the official EDTA repository:

`https://github.com/macjiffriya/EDTA.git`

Our own implementation and experimental code are available at:

`https://github.com/nonit-gupta/Drug-Target-Affinity-Prediction`